

COM3031-Probabilistic Machine Learning Workshop

Importance, Rejection Sampling, and MCMC

Overview

This workshop focuses on three sampling methods: importance sampling, rejection sampling, and Markov Chain Monte Carlo (MCMC) via Metropolis–Hastings (MH). The goal is to implement these methods to approximate a target distribution. When direct sampling from the target distribution $p(x)$ is difficult, we can use a proposal distribution $q(x)$ to help: importance sampling reweights samples from $q(x)$, rejection sampling accepts/rejects samples from $q(x)$, and MCMC constructs a Markov chain whose stationary distribution is $p(x)$.

You will be using the Jupyter notebook provided on ELE to complete the following tasks. You may need to go through the lecture slides/notes of Week 5. The solutions will be published by this week.

Tasks

1. Implement importance sampling by filling in the missing code in the Jupyter notebook.
2. Repeat the above step with a different proposal distribution and observe the effect on the approximation.
3. Implement rejection sampling and calculate the acceptance ratio by filling in the missing code in the Jupyter notebook.
4. Investigate the effect of using different values of k on the acceptance ratio.
5. Investigate the effect of using different total numbers of samples on the approximation of the target distribution.
6. Implement Markov Chain Monte Carlo (MCMC) using the Metropolis–Hastings (MH) algorithm in PyMC (i.e., the `Metropolis` step method) to draw samples from the target distribution (up to a normalizing constant).
7. Plot the MH trace and a histogram of MH samples (after burn-in) overlaid with the target density. Report the acceptance rate.
8. Try to compare importance sampling, rejection sampling, and MH (MCMC) on the same target distribution.